

Two-Pump Distributed Parametric Amplification of 25x 32Gbaud 16-QAM Channels over 25 km SSMF with ~20dB Gain

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Abstract We demonstrate distributed fiber optic parametric amplification of 25x16-QAM 32Gbaud channels over 25 km of G.652.D in the O-band with the GMI-estimated throughput of 3.148Tbps and on/off gain of ~20dB by employing two ~125mW pumps. ©2026 The Author(s)

Introduction

Distributed amplification plays a crucial role in modern optical communications facilitating high throughput and reach [1], [2]. Commonly, distributed amplification relies on stimulated Raman scattering of a transmission fiber [3]. Recently, it has been demonstrated that distributed parametric amplification relying on four-wave mixing in standard single mode fiber (SSMF) can provide high gain (>10 dB) whilst employing low pump powers (<200 mW) [4], [5]. Indeed, parametric gain coefficient is ~5 times higher than Raman gain coefficient [6], and unidirectional parametric gain is less prone to double Rayleigh scattering than Raman amplifiers [7]. Furthermore, distributed parametric amplification can be also employed in

phase-sensitive mode providing virtually noise-less amplification [8]. However, demonstrations of high-gain distributed parametric amplification in SSMF in [4] and [5] have been limited to amplification of only one and two 50 Gbps PAM4 channels.

In this paper, we perform distributed parametric amplification of an equivalent of 25x 16-QAM 32 Gbaud single-polarization channels and achieve the record generalized mutual information (GMI)-estimated throughput of 3.148 Tbps for distributed parametric amplification. This has been achieved by designing a distributed parametric amplifier with ~20 dB on/off gain whilst employing only two 21dBm pumps (at the input of the transmission fiber). We demonstrate that even higher gain, up to 27 dB is possible with

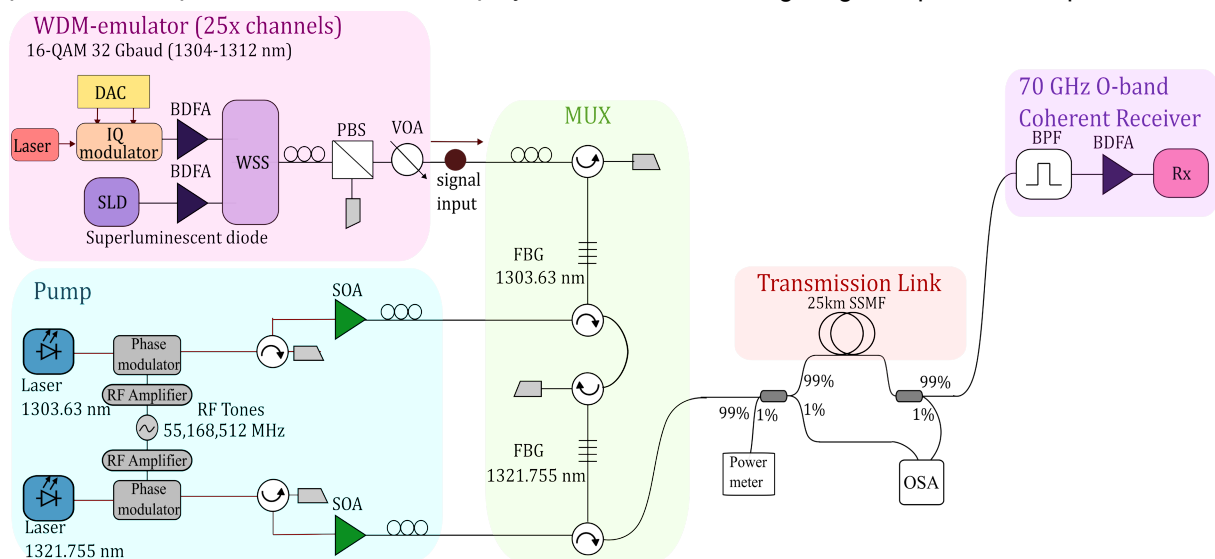


Fig. 1. Experimental setup for distributed parametric amplification of equivalent of 25x 16-QAM 32Gbaud channels in 25km SSMF. WDM – wavelength division multiplexing, SLD – super luminescent diode, DAC – digital to analog converter, BDFA – Bismuth-doped fiber amplifier, MUX – multiplexer, PBS – polarization beam splitter, VOA – variable optical attenuator, FBG – fiber Bragg grating, SOA – semiconductor optical amplifier, SSMF – standard single mode fiber, OSA – optical spectrum analyzer, BPF – band pass filter, Rx – receiver.

slightly higher pump powers of 160mW. Consequently, this work confirms a promising potential of distributed parametric amplification for power-efficient amplification in high-capacity links in the O band.

Experimental setup

Experimental setup of the two-pump distributed FOPA is shown in Fig. 1. The WDM emulator produced an equivalent of 25x 16-QAM 32 GBaud channels. Then, the channels were combined with two pumps by a multiplexer (MUX) assembly, and altogether they were transmitted over a 25 km SSMF link. Signals were amplified along the link and detected by a coherent receiver.

The WDM signal emulator produced a combination of a single-polarization 16-QAM 32 GBaud signal tuneable to each of 25x50 GHz-spaced frequencies in the range of 1304.77-1311.62 nm, and white noise emulating the remaining channels in this wavelength range. The signal was sourced from an external cavity laser tunable in the O-band and modulated with an IQ modulator. The white noise for emulated channels was sourced from a super luminescent diode (SLD). Then, the signal and the white noise were each amplified by bismuth-doped fiber amplifiers (BDFAs) and combined by a wavelength selective switch (WSS). Additionally, the WSS was used to shape the noise and limited its wavelength range to the emulated WDM range. Then, the signal and the white noise were passed through a polarization beam splitter (PBS) to ensure they are co-polarized because as only single-polarization parametric amplification was investigated in this work. A variable optical attenuator (VOA) at the output of the WDM emulator adjusted output power of the emulated WDM channels. Fig. 2 shows the optical power spectrum at the WDM emulator output. The signal power at that point ('Signal input' in Fig. 1) was -23 dBm and the total power was -8 dBm.

Each of the two pumps for parametric amplification were sourced from an external-cavity continuous-wave (CW) tunable O-band laser with linewidth of ~100 kHz. The pump wavelengths were 1303.630 nm and 1321.755 nm. Each pump was phase modulated with three RF tones at 55, 168 and 512 MHz to mitigate stimulated Brillouin scattering [9]. The tones' respective delays were adjusted so the pumps' phase modulation appears in counter-phase in the SSMF, so its impact on amplified signals is minimized [10]. Then, each pump was amplified by an SOA with a maximum output power of ~26 dBm. The pumps' polarization was

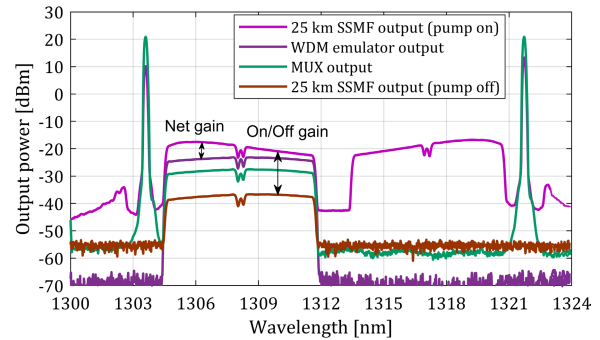


Fig. 2: Example optical power spectra throughout the setup when pump powers at the MUX output are 21dBm, and the signal wavelength is 1308.1nm.

adjusted to co-polarize them, and then pumps were combined with the WDM signals by a MUX.

The MUX assembly combined the emulated WDM and the two pumps by means of two fiber Bragg gratings (FBGs) tuned to the pump wavelengths' and four circulators. The MUX insertion loss for the WDM signals was ~6 dB, and for the pumps ~4 dB and ~2 dB respectively. Optical power spectrum at the MUX output is shown in Fig. 2.

Then, the WDM signals and the pumps were injected into 25 km of G.652.D fiber, which had an average loss of ~9 dB. The input signal power was -29 dBm per channel at this point. Calibrated 1% tap couplers were employed at the input and output of the transmission link to monitor optical power spectra.

Output from the link was passed to the receiver. The bandpass filter (BPF) was used to select the channel under test. The filtered signal was amplified by a BDFA, detected by a coherent receiver and processed with offline digital signal processing (DSP).

Results and Discussion

Fig. 2 shows an example of an optical spectra with each pump power at the input of the link being 21 dBm. The 25 km SSMF output with pumps 'on' also shows generation of idlers in the range 1314...1321 nm. The difference between the 25km SSMF output with pumps on (off) and the WDM emulator output was the net gain (overall insertion loss). The latter was ~16 dB, and it was added to the net gain to estimate on/off gain.

Fig. 3(a) shows net gain (left axis) and the estimated on/off gain (right axis) spectra of parametric amplification in the 25 km SSMF link for three pump power levels: 20.3 dBm, 21 dBm and 22 dBm per pump at the input of the SSMF. All cases deliver at least 16 dB on/off gain and a positive net gain in the 25 km link. A small gain tilt from of 2...5 dB is observed across the gain spectra. This gain tilt can be reduced by adjusting pump wavelengths within ~0.02 nm. At the

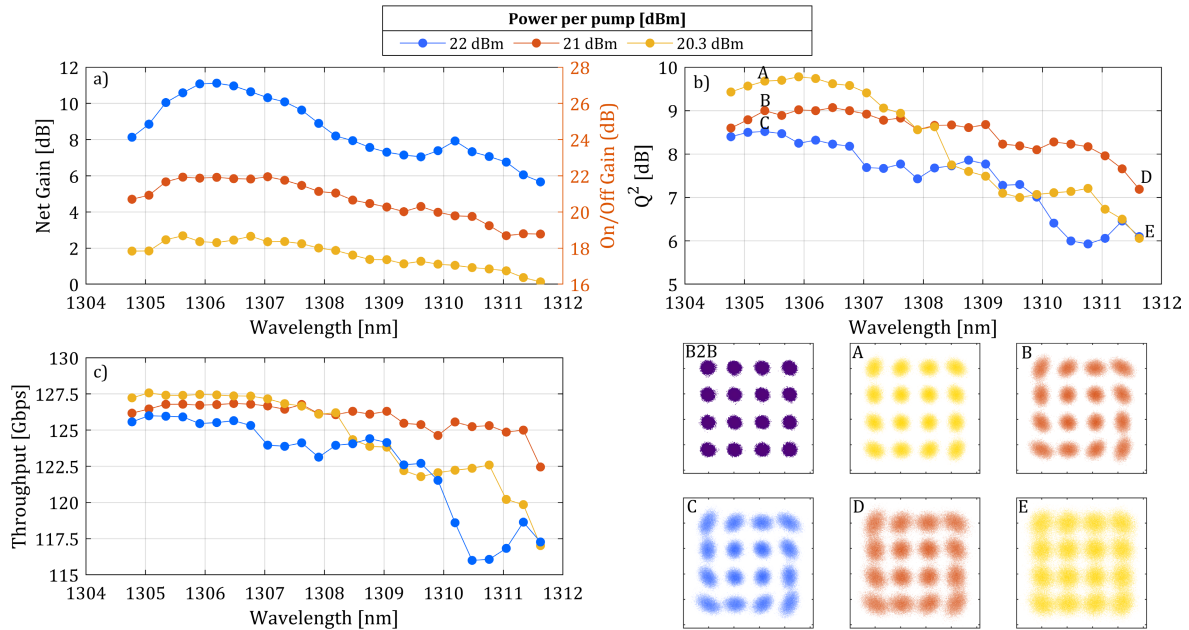


Fig. 3. (a) Net and on/off gain, (b) Q^2 -factor, and (c) throughput per channel for each of 25 signal wavelengths for three pump power levels of 20.3 dBm, 21 dBm and 22 dBm. Constellation diagrams are shown for the B2B case and 5 measurements denoted by letters A-E at the plot (b).

highest pumping level of 22 dBm (160 mW) per pump, on/off gain is 22-27 dB, and net gain is 6-11 dB. Such high gain values might enable 'lossless' spans up to 50 km long in the O-band.

Fig. 3(b) shows the measured Q^2 -factor (Q^2_{dB}) for the 16 QAM channel as a function of wavelength. The Q^2 -factor was obtained from the bit error ratio (BER) averaged across eight 10 μ s traces captured for each point. On the short wavelength side the pump power increase degrades the Q^2 -factor. Constellations for the points A, B, and C demonstrate that phase noise is aggravated in case of higher pump powers. Likely, that is due to nonlinear signal distortion and/or residual pump phase modulation transfer both linked to the signal gain. On the long wavelength side, the Q^2 -factor rolls off for all curves, especially for the curve corresponding to 20.3 dBm pump power. Constellations for the points D and E demonstrate a higher level of white noise for these channels, so this might be due to insufficient gain in this part of the spectrum.

The system performance was evaluated by estimating the achievable data rate using generalized mutual information (GMI). Fig. 3(c) presents the wavelength-dependent GMI-estimated data rate, obtained by averaging over eight 10 μ s acquisitions, corresponding to a total of 2.56×10^6 symbols.

The highest overall throughput of 3.148 Tbps was estimated based on the GMI with the power per pump of 21 dBm. Indeed, this pumping level provided the most balanced Q^2 -factor across the spectrum, trading off between phase noise on the

short wavelength side and white noise on the long wavelength side.

We used an additive white Gaussian noise (AWGN)-based GMI model to estimate Q^2 -factor and throughput. While this approach is not strictly accurate in the presence of observed phase noise, the resulting inaccuracy is negligible for the purposes of this work, which is to demonstrate the capability of distributed parametric amplification to support WDM QAM transmission.

Conclusion

We have demonstrated capability of distributed parametric amplification to provide high on/off gain of ~20 dB in the wavelength range 1304-1312 nm and to amplify an equivalent of 25x 16-QAM 32 GBaud single-polarization channels with total GMI-estimated throughput of 3.148 Tbps, employing two pumps with power of 21 dBm each in a 25 km length SSMF. Consequently, distributed parametric amplification might be a power-efficient alternative to distributed Raman amplification. Furthermore, positive net gain after a 25 km link implies possibility of 'lossless' transmission over spans up to 50 km long in the O-band.

Acknowledgements

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